

EXHIBIT 31

EXHIBIT 31

U.S. Patent No. 8,507,983 Infringement Chart

Title: High voltage device

Inventors: Guowei Zhang, Purakh Raj Verma

Abstract: A device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. A doped drift well adjacent to a second edge of the gate is also included.

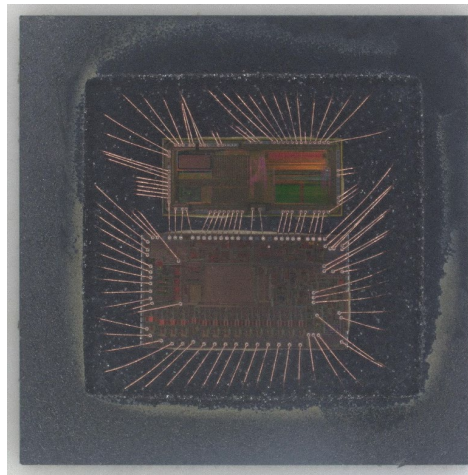
Claim	Tower '983 Accused Products
[1pre] A device comprising:	The '983 Accused Tower Products include products manufactured using Tower's BCD processes (including its TS18SOI process). ¹ For example, Tower manufactures the PA22BZ die in the Qorvo PAC22140 using a 180nm BCD process referred to by Tower as TS18SOI. To the extent the preamble is limiting, the Tower '983 Accused Products comprise a device, as explained further in the following limitations.

¹ GlobalFoundries reserves the right to identify different or additional Tower '983 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

EXHIBIT 31



On information and belief, the PA22BZ (bottom of image below) has substantially similar structure, function, operation, and implementation with respect to the claims of the '983 patent as any other Tower '983 Accused Products.



For example, a cross-sectional image of the PA22BZ die (as illustrated by the red line in the image below) reveals a structure that is representative of the infringing functionality in the '983 Tower Accused Products.

EXHIBIT 31

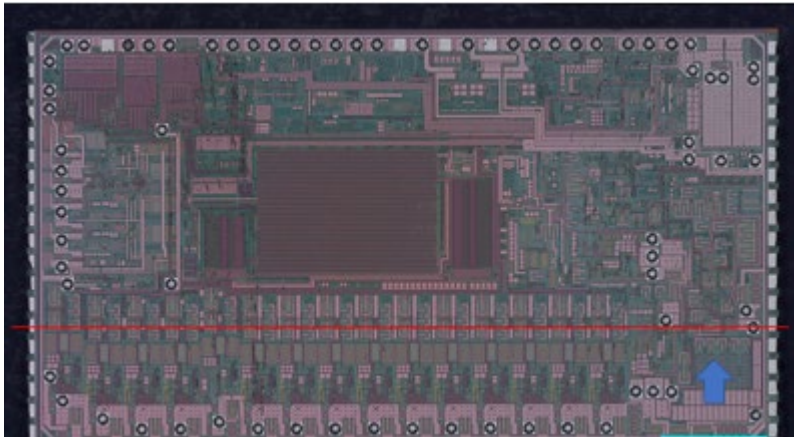
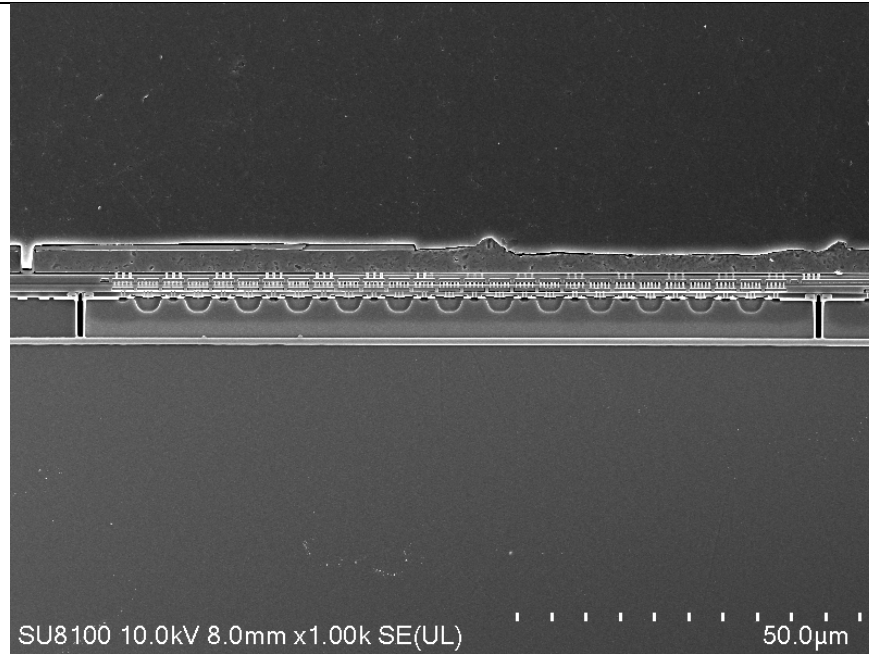
	 A micrograph of a semiconductor device, likely a microprocessor, showing a complex circuit layout. A horizontal red line is drawn across the center of the image, and a blue arrow points upwards from the bottom edge towards the center.
[1a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;	The Tower '983 Accused Products comprise a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric. For example, the Tower '983 Accused Products include a substrate (bottom of image below).

EXHIBIT 31



The active device region in the Tower '983 Accused Products includes a gate that has a gate electrode over a gate dielectric, as shown in the reverse engineering analysis below.

EXHIBIT 31

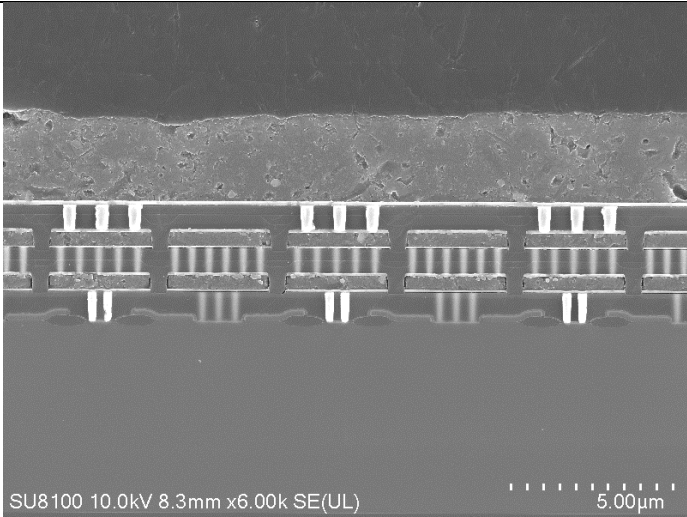
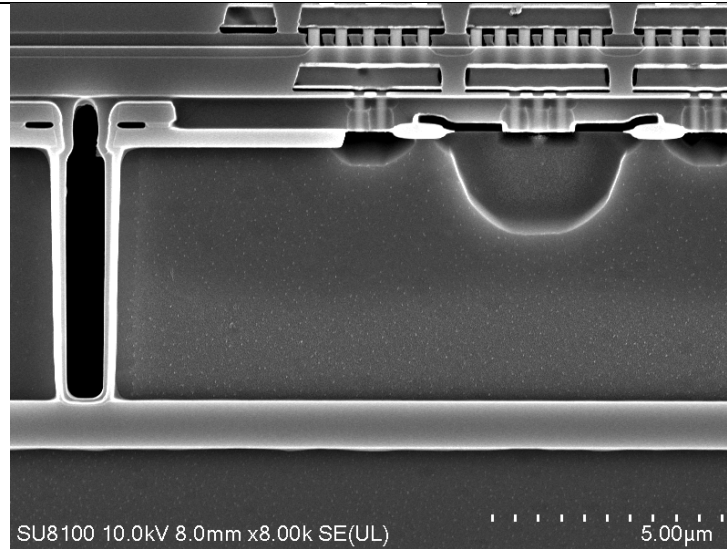
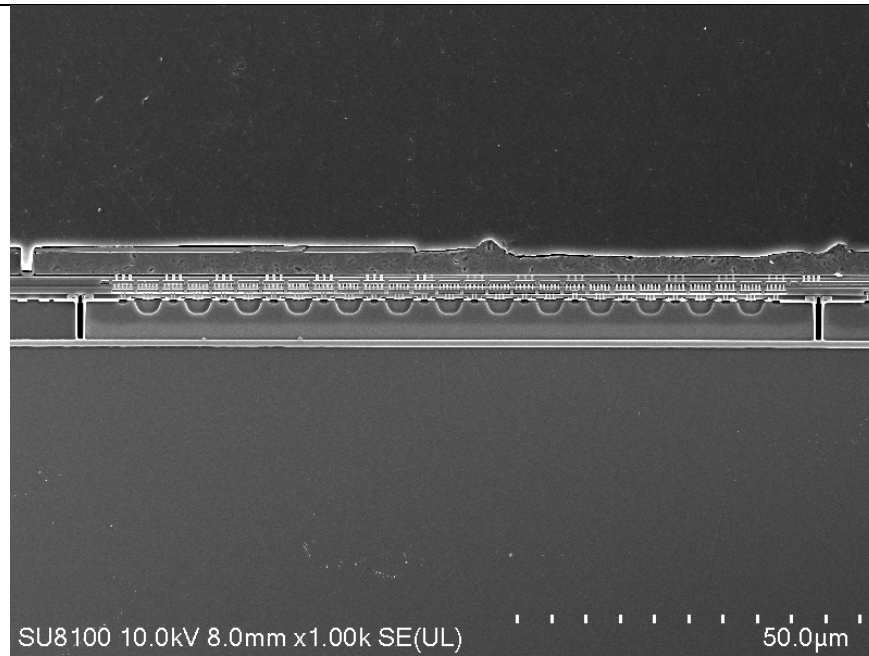
	
[1b] a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and	The Tower '983 Accused Products comprise a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. For example, the reverse engineering analysis below illustrates a doped channel well that is overlapped by a first edge of the gate to form a channel.

EXHIBIT 31



The channel is displaced from a heavily doped region that is adjacent to the edge of the gate.

EXHIBIT 31



For example, the SCM analysis below shows the doping profile for the active device region in the TEM image above.

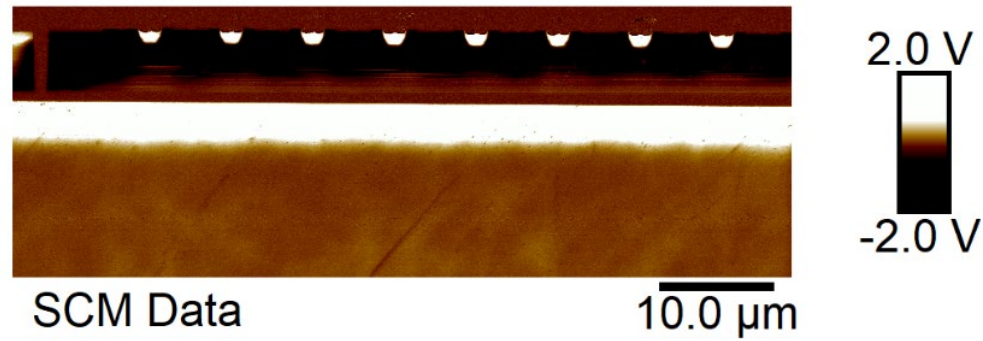


EXHIBIT 31

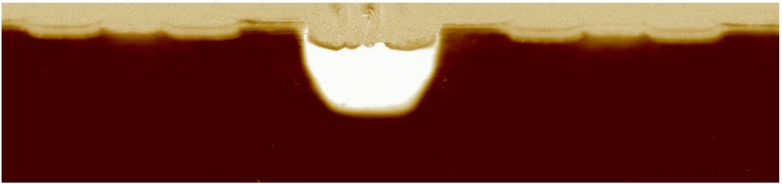
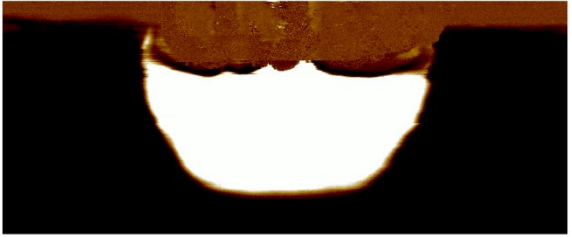
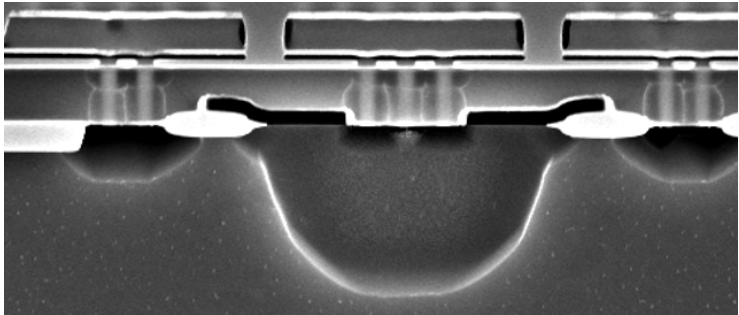
	The SCM analysis below shows a magnified view of the doping profiles in the above image.  SCM Data 2.0 V -2.0 V 2.0 μm  SCM Data 2.0 V -2.0 V 780.0 nm
[1c] a doped drift well adjacent to a second edge of the gate.	The Tower '983 Accused Products comprise a doped drift well adjacent to a second edge of the gate. For example, the image shows a doped drift well adjacent to a second edge of the gate. 

EXHIBIT 31

[13pre] A device comprising:	The Tower '983 Accused Products comprise a device. <i>See [1pre]</i> .
[13a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;	The Tower '983 Accused Products comprise a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric. <i>See [1a]</i> .
[13b] a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and	The Tower '983 Accused Products comprise a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate. <i>See [1b]</i> .
[13c] a doped drift well adjacent to a second edge of the gate; and	The Tower '983 Accused Products comprise a doped drift well adjacent to a second edge of the gate. <i>See [1c]</i> .

EXHIBIT 31

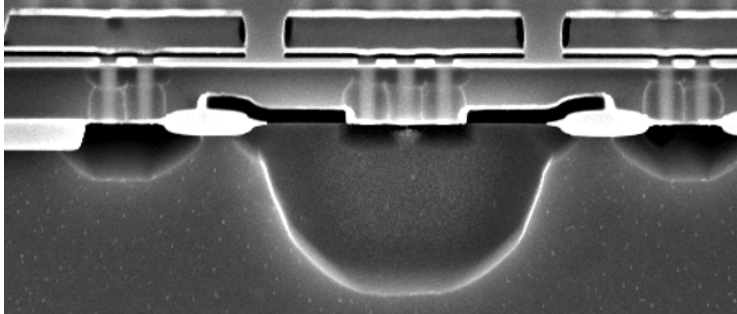
[13d] a drift isolation region in the doped drift well.	The Tower '983 Accused Products comprise a drift isolation region in the doped drift well that is adjacent to a second edge of the gate, as shown in the images below. 
[20pre] A semiconductor device comprising:	The Tower '983 Accused Products comprise a semiconductor device. <i>See [1pre].</i>
[20a] a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;	The Tower '983 Accused Products comprise a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric. <i>See [1a].</i>
[20b] first and second heavily doped regions adjacent to first and second edges of the gate;	The Tower '983 Accused Products comprise first and second heavily doped regions adjacent to first and second edges of the gate.

EXHIBIT 31

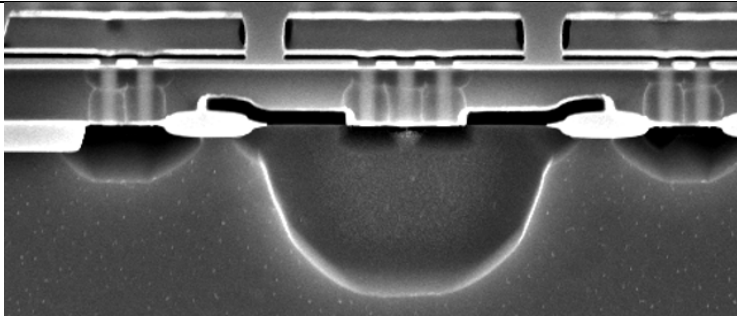
	
[20c] a doped channel well disposed in the substrate adjacent to the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from the first heavily doped region adjacent to the first edge of the gate; and	The Tower '983 Accused Products comprise a doped channel well disposed in the substrate adjacent to the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from the first heavily doped region adjacent to the first edge of the gate. <i>See [1c].</i>
[20d] a doped drift well adjacent to a second edge of the gate, wherein a drift isolation	The Tower '983 Accused Products comprise a doped drift well adjacent to a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well. <i>See [1c] and 13[d].</i>

EXHIBIT 31

region is disposed in the doped drift well.	
--	--